THE FITTING-TYPE IDEALS OF A BOUNDED COMPLEX OF FINITE FREE MODULES BASE CHANGE AS EXPECTED WHEN A NONZERODIVISOR MAPS TO A NONZERODIVISOR

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Allowed standard prerequisites (editorial).

- Fitting-type ideals I_i(M^bullet, f) of a bounded complex of finite free modules. the notation I_i in the statement at source lines 28016-28017. Fixes the two ideals being compared and the augmented maps whose minors generate them.
- Base change of a complex of finite free modules and of the minors of a matrix. used at proof lines 28021-28023. Identifies the generators of the two ideals term by term, which is the whole content of the proof.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ be a bounded complex of finite free A-modules. Assume f maps to a nonzerodivisor g in B. Then $I_i(M^\bullet, f)B = I_i(M^\bullet \otimes_A B, g)$.*

Proof. The minors of $(f, d^i) : M^i \to M^i \oplus M^{i+1}$ map to the corresponding minors of $(g, d^i) : M^i \otimes_A B \to M^i \otimes_A B \oplus M^{i+1} \otimes_A B$. □

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